D Concentration Inequalities

In this appendix, we present several *concentration inequalities* used in the proofs given in this book. Concentration inequalities give probability bounds for a random variable to be concentrated around its mean, or for it to deviate from its mean or some other value.

D.1 Hoeffding's inequality

We first present Hoeffding's inequality, whose proof makes use of the general *Chernoff bounding technique*. Given a random variable X and $\epsilon > 0$, this technique consists of proceeding as follows to bound $\mathbb{P}[X \geq \epsilon]$. For any $t > 0$, first Markov's inequality is used to bound $\mathbb{P}[X \geq \epsilon]$:

$$\mathbb{P}[X \geq \epsilon] = \mathbb{P}[e^{tX} \geq e^{t\epsilon}] \leq e^{-t\epsilon}\, \mathbb{E}[e^{tX}]\,. \tag{D.1}$$

Then, an upper bound $g(t)$ is found for $\mathbb{E}[e^{tX}]$ and t is selected to minimize $e^{-t\epsilon}g(t)$. For Hoeffding's inequality, the following lemma provides an upper bound on $\mathbb{E}[e^{tX}]$.

Lemma D.1 (Hoeffding's lemma) *Let X be a random variable with $E[X] = 0$ and $a \leq X \leq b$ with $b > a$. Then, for any $t > 0$, the following inequality holds:*

$$\mathbb{E}[e^{tX}] \leq e^{\frac{t^2(b-a)^2}{8}}\,. \tag{D.2}$$

Proof: By the convexity of $x \mapsto e^x$, for all $x \in [a, b]$, the following holds:

$$e^{tx} \leq \frac{b-x}{b-a}e^{ta} + \frac{x-a}{b-a}e^{tb}\,.$$

Thus, using $\mathbb{E}[X] = 0$,

$$\mathbb{E}[e^{tX}] \leq \mathbb{E}\left[\frac{b-X}{b-a}e^{ta} + \frac{X-a}{b-a}e^{tb}\right] = \frac{b}{b-a}e^{ta} + \frac{-a}{b-a}e^{tb} = e^{\phi(t)}\,,$$

where,

$$\phi(t) = \log\left(\frac{b}{b-a}e^{ta} + \frac{-a}{b-a}e^{tb}\right) = ta + \log\left(\frac{b}{b-a} + \frac{-a}{b-a}e^{t(b-a)}\right)\,.$$

For any $t > 0$, the first and second derivative of ϕ are given below:

$$\begin{aligned}
\phi'(t) &= a - \frac{ae^{t(b-a)}}{\frac{b}{b-a} - \frac{a}{b-a}e^{t(b-a)}} = a - \frac{a}{\frac{b}{b-a}e^{-t(b-a)} - \frac{a}{b-a}}\,,\\
\phi''(t) &= \frac{-abe^{-t(b-a)}}{[\frac{b}{b-a}e^{-t(b-a)} - \frac{a}{b-a}]^2}\\
&= \frac{\alpha(1-\alpha)e^{-t(b-a)}(b-a)^2}{[(1-\alpha)e^{-t(b-a)} + \alpha]^2}\\
&= \frac{\alpha}{[(1-\alpha)e^{-t(b-a)} + \alpha]}\frac{(1-\alpha)e^{-t(b-a)}}{[(1-\alpha)e^{-t(b-a)} + \alpha]}(b-a)^2\,.
\end{aligned}$$

where α denotes $\frac{-a}{b-a}$. Note that $\phi(0) = \phi'(0) = 0$ and that $\phi''(t) = u(1-u)(b-a)^2$ where $u = \frac{\alpha}{[(1-\alpha)e^{-t(b-a)}+\alpha]}$. Since u is in $[0,1]$, $u(1-u)$ is upper bounded by $1/4$ and $\phi''(t) \leq \frac{(b-a)^2}{4}$. Thus, by the second order expansion of function ϕ, there exists $\theta \in [0,t]$ such that:

$$\phi(t) = \phi(0) + t\phi'(0) + \frac{t^2}{2}\phi''(\theta) \leq t^2 \frac{(b-a)^2}{8}\,, \tag{D.3}$$

which completes the proof. □

The lemma can be used to prove the following result known as *Hoeffding's inequality*.

Theorem D.2 (Hoeffding's inequality) *Let $X_1, \ldots, X_m$ be independent random variables with X_i taking values in $[a_i, b_i]$ for all $i \in [m]$. Then, for any $\epsilon > 0$, the following inequalities hold for $S_m = \sum_{i=1}^m X_i$:*

$$\mathbb{P}[S_m - \mathbb{E}[S_m] \geq \epsilon] \leq e^{-2\epsilon^2/\sum_{i=1}^m (b_i-a_i)^2} \tag{D.4}$$

$$\mathbb{P}[S_m - \mathbb{E}[S_m] \leq -\epsilon] \leq e^{-2\epsilon^2/\sum_{i=1}^m (b_i-a_i)^2}\,. \tag{D.5}$$

Proof: Using the Chernoff bounding technique and lemma D.1, we can write:

$$\begin{aligned}
\mathbb{P}[S_m - \mathbb{E}[S_m] \geq \epsilon] &\leq e^{-t\epsilon}\,\mathbb{E}[e^{t(S_m - \mathbb{E}[S_m])}] \\
&= e^{-t\epsilon}\Pi_{i=1}^m\,\mathbb{E}[e^{t(X_i - \mathbb{E}[X_i])}] && \text{(independence of } X_i\text{s)} \\
&\leq e^{-t\epsilon}\Pi_{i=1}^m e^{t^2(b_i-a_i)^2/8} && \text{(lemma D.1)} \\
&= e^{-t\epsilon} e^{t^2\sum_{i=1}^m (b_i-a_i)^2/8} \\
&\leq e^{-2\epsilon^2/\sum_{i=1}^m (b_i-a_i)^2}\,,
\end{aligned}$$

where we chose $t = 4\epsilon/\sum_{i=1}^m (b_i - a_i)^2$ to minimize the upper bound. This proves the first statement of the theorem, and the second statement is shown in a similar way. □

When the variance $\sigma^2_{X_i}$ of each random variable X_i is known and the $\sigma^2_{X_i}$s are relatively small, better concentration bounds can be derived (see *Bennett's* and *Bernstein's inequalities* proven in exercise D.6).

D.2 Sanov's theorem

Here, we present a finer upper bound than Hoeffding's inequality expressed in terms of the binary relative entropy.

Theorem D.3 (Sanov's theorem) *Let $X_1, \ldots, X_m$ be independent random variables drawn according to some distribution $\mathcal{D}$ with mean p and support included in $[0,1]$. Then, for any $q \in [0,1]$, the following inequality holds for $\widehat{p} = \frac{1}{m}\sum_{i=1}^m X_i$:*

$$\mathbb{P}[\widehat{p} \geq q] \leq e^{-m\mathsf{D}(q\|p)}\,,$$

where $\mathsf{D}(q\|p) = q\log\frac{q}{p} + (1-q)\log\frac{1-q}{1-p}$ is the binary relative entropy of p and q.

Proof: For any $t > 0$, by convexity of the function $x \mapsto e^{tx}$, the following inequality holds for all $x \in [0,1]$: $e^{tx} = e^{t[(1-x)\cdot 0 + x\cdot 1]} \leq 1 - x + e^t x$. In view of that, for any $t > 0$, we can write

$$\begin{aligned}
\mathbb{P}[\widehat{p} \geq q] &= \mathbb{P}[e^{tm\widehat{p}} \geq e^{tmq}] \\
&= \mathbb{P}[e^{tm\widehat{p}} \geq e^{tmq}] \\
&\leq e^{-tmq}\, \mathbb{E}[e^{tm\widehat{p}}] && \text{(by Markov's inequality)} \\
&= e^{-tmq}\, \mathbb{E}[e^{t\sum_{i=1}^m X_i}] \\
&= e^{-tmq} \prod_{i=1}^m \mathbb{E}[e^{tX_i}] \\
&\leq e^{-tmq} \prod_{i=1}^m \mathbb{E}[1 - X_i + e^t X_i] && (\forall x \in [0,1], e^{tx} \leq 1 - x + e^t x) \\
&= [e^{-tq}(1 - p + e^t p)]^m.
\end{aligned}$$

Now, the function $f\colon t \mapsto e^{-tq}(1-p+e^t p) = (1-p)e^{-tq} + pe^{t(1-q)}$ reaches its minimum at $t = \log \frac{q(1-p)}{p(1-q)}$. Plugging in this value of t in the inequality above yields $\mathbb{P}[\widehat{p} \geq q] \leq e^{-m\mathsf{D}(q\|p)}$. □

Note that for any $\epsilon > 0$, $\epsilon \leq 1 - p$, with the choice $q = p + \epsilon$, the theorem implies

$$\mathbb{P}[\widehat{p} \geq p + \epsilon] \leq e^{-m\mathsf{D}(p+\epsilon\|p)}\,. \tag{D.6}$$

This is a finer bound than Hoeffding's inequality (Theorem D.2) since, by Pinsker's inequality (Proposition E.7), $\mathsf{D}(p+\epsilon\|p) \geq \frac{1}{2}(2\epsilon)^2 = 2\epsilon^2$. Similarly, we can derive a symmetric bound by applying the theorem to the random variables $Y_i = 1 - X_i$. Then, for any $\epsilon > 0$, $\epsilon \leq p$, with the choice $q = p - \epsilon$, the theorem implies

$$\mathbb{P}[\widehat{p} \leq p - \epsilon] \leq e^{-m\mathsf{D}(p-\epsilon\|p)}\,. \tag{D.7}$$

D.3 Multiplicative Chernoff bounds

Sanov's theorem can be used to prove the following *multiplicative Chernoff bounds*.

Theorem D.4 (Multiplicative Chernoff bounds) *Let $X_1, \ldots, X_m$ be independent random variables drawn according to some distribution $\mathcal{D}$ with mean p and support included in $[0,1]$. Then, for any $\gamma \in \left[0, \frac{1}{p} - 1\right]$, the following inequality holds for $\widehat{p} = \frac{1}{m}\sum_{i=1}^m X_i$:*

$$\mathbb{P}[\widehat{p} \geq (1+\gamma)p] \leq e^{-\frac{mp\gamma^2}{3}}$$

$$\mathbb{P}[\widehat{p} \leq (1-\gamma)p] \leq e^{-\frac{mp\gamma^2}{2}}\,.$$

Proof: The proof consists of deriving in each case a finer lower bound for the binary relative entropy than Pinsker's inequality. Using the inequalities $\log(1+x) \geq \frac{x}{1+\frac{x}{2}}$ and $\log(1+x) < x$, we can write

$$\begin{aligned}
-\mathsf{D}((1+\gamma)p\|p) &= (1+\gamma)p \log \frac{p}{(1+\gamma)p} + (1-(1+\gamma)p)\log\left[\frac{1-p}{1-(1+\gamma)p}\right] \\
&= (1+\gamma)p \log \frac{1}{1+\gamma} + (1-p-\gamma p)\log\left[1 + \frac{\gamma p}{1-p-\gamma p}\right] \\
&\leq (1+\gamma)p \frac{-\gamma}{1+\frac{\gamma}{2}} + (1-p-\gamma p)\frac{\gamma p}{1-p-\gamma p} \\
&= \gamma p\left[1 - \frac{1+\gamma}{1+\frac{\gamma}{2}}\right] = \frac{-\frac{\gamma^2 p}{2}}{1+\frac{\gamma}{2}} = \frac{-\gamma^2 p}{2+\gamma} \\
&\leq \frac{-\gamma^2 p}{2+1} = \frac{-\gamma^2 p}{3}.
\end{aligned}$$

Similarly, using the inequalities $(1-x)\log(1-x) \geq -x+\frac{x^2}{2}$ valid for $x \in (0,1)$ and $\log(1-x) < -x$, we can write

$$\begin{aligned}
-\mathsf{D}((1-\gamma)p\|p) &= (1-\gamma)p\log\frac{p}{(1-\gamma)p} + (1-(1-\gamma)p)\log\left[\frac{1-p}{1-(1-\gamma)p}\right] \\
&= (1-\gamma)p\log\frac{1}{1-\gamma} + (1-p+\gamma p)\log\left[1-\frac{\gamma p}{1-p+\gamma p}\right] \\
&\leq \left(\gamma-\frac{\gamma^2}{2}\right)p + (1-p+\gamma p)\frac{-\gamma p}{1-p+\gamma p} = \frac{-\gamma^2 p}{2}.
\end{aligned}$$

This completes the proof. □

D.4 Binomial distribution tails: Upper bounds

Let $X_1, \ldots, X_m$ be independent random variables taking values in $\{0,1\}$ with $\mathbb{P}[X_i = 1] = p \in [0,1]$ for $i = 1, \ldots, m$. Then, $\sum_{i=1}^m X_i$ follows the binomial distribution $B(m,p)$. We will denote by $\overline{X}$ the average $\overline{X} = \frac{1}{m}\sum_{i=1}^m X_i$. Then, the following equality and inequalities hold:

$$\mathbb{P}\left[\overline{X} - p > \epsilon\right] = \sum_{k=\lceil (p+\epsilon)m \rceil}^{m} \binom{m}{k} p^k (1-p)^{m-k} \qquad \text{(Binomial formula)}$$

$$\mathbb{P}\left[\overline{X} - p > \epsilon\right] \leq e^{-2m\epsilon^2} \qquad \text{(Hoeffding's inequality)}$$

$$\mathbb{P}\left[\overline{X} - p > \epsilon\right] \leq e^{-\frac{m\epsilon^2}{2\sigma^2+\frac{2\epsilon}{3}}} \qquad \text{(Bernstein's inequality)}$$

$$\mathbb{P}\left[\overline{X} - p > \epsilon\right] \leq e^{-m\sigma^2\theta\left(\frac{\epsilon}{\sigma^2}\right)} \qquad \text{(Bennett's inequality)}$$

$$\mathbb{P}\left[\overline{X} - p > \epsilon\right] \leq e^{-m\mathsf{D}(p+\epsilon\|p)} \qquad \text{(Sanov's inequality),}$$

where $\sigma^2 = p(1-p) = \mathrm{Var}[X_i]$ and $\theta(x) = (1+x)\log(1+x) - x$. The last three inequalities are shown in exercises D.6 and D.7. Using Bernstein's inequality, for example, we can see that for ϵ relatively small, that is $\epsilon \ll 2\sigma^2$, the upper bound is approximately of the form $e^{-\frac{m\epsilon^2}{2\sigma^2}}$ and thus admits a Gaussian behavior. For $\epsilon \gg 2\sigma^2$, $e^{-\frac{3m\epsilon}{2}}$, the upper bound admits a Poisson behavior.

Figure D.1 shows a comparison of these bounds for different values of the variance $\sigma^2 = p(1-p)$: small variance ($p = .05$), large variance ($p = .5$).

D.5 Binomial distribution tails: Lower bound

Let X be a random variable following the binomial distribution $B(m,p)$ and let k be an integer such that $p \leq \frac{1}{4}$ and $k \geq mp$ or $p \leq \frac{1}{2}$ and $mp \leq k \leq m(1-p)$. Then, the following inequality known as *Slud's inequality* holds:

$$\mathbb{P}[X \geq k] \geq \mathbb{P}\left[N \geq \frac{k-mp}{\sqrt{mp(1-p)}}\right], \tag{D.8}$$

where N is in standard normal form.

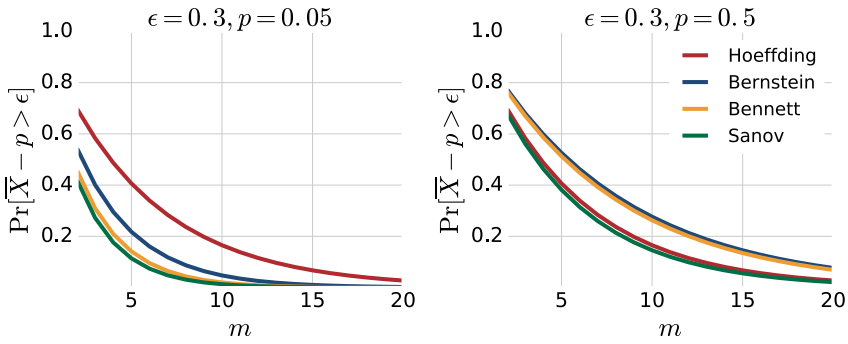

Figure D.1
Comparison of tail bounds for a binomial random variable for $\epsilon = .3$ and $p = .05$ (small variance) or $p = .5$ (maximal variance) as a function of the sample size m.

D.6 Azuma's inequality

This section presents a concentration inequality that is more general than Hoeffding's inequality. Its proof makes use of a Hoeffding's inequality for *martingale differences*.

Definition D.5 (Martingale difference) *A sequence of random variables $V_1, V_2, \ldots$ is a martingale difference sequence with respect to $X_1, X_2, \ldots$ if for all $i > 0$, V_i is a function of $X_1, \ldots, X_i$ and*

$$\mathbb{E}[V_{i+1} | X_1, \ldots, X_i] = 0\,. \tag{D.9}$$

The following result is similar to Hoeffding's lemma.

Lemma D.6 *Let V and Z be random variables satisfying $\mathbb{E}[V|Z] = 0$ and, for some function f and constant $c \geq 0$, the inequalities:*

$$f(Z) \leq V \leq f(Z) + c\,. \tag{D.10}$$

Then, for all $t > 0$, the following upper bound holds:

$$\mathbb{E}[e^{tV} | Z] \leq e^{t^2 c^2/8}\,. \tag{D.11}$$

Proof: The proof follows using the same steps as in that of lemma D.1 with conditional expectations used instead of expectations: conditioned on Z, V takes values in $[a, b]$ with $a = f(Z)$ and $b = f(Z) + c$ and its expectation vanishes. □

The lemma is used to prove the following theorem, which is one of the main results of this section.

Theorem D.7 (Azuma's inequality) *Let $V_1, V_2, \ldots$ be a martingale difference sequence with respect to the random variables $X_1, X_2, \ldots$, and assume that for all $i > 0$ there is a constant $c_i \geq 0$ and random variable Z_i, which is a function of $X_1, \ldots, X_{i-1}$, that satisfy*

$$Z_i \leq V_i \leq Z_i + c_i\,. \tag{D.12}$$

Then, for all $\epsilon > 0$ and m, the following inequalities hold:

$$\mathbb{P}\left[\sum_{i=1}^{m} V_i \geq \epsilon\right] \leq \exp\left(\frac{-2\epsilon^2}{\sum_{i=1}^{m} c_i^2}\right) \tag{D.13}$$

$$\mathbb{P}\left[\sum_{i=1}^{m} V_i \leq -\epsilon\right] \leq \exp\left(\frac{-2\epsilon^2}{\sum_{i=1}^{m} c_i^2}\right)\,. \tag{D.14}$$

Proof: For any $k \in [m]$, let $S_k = \sum_{i=1}^k V_i$. Then, using Chernoff's bounding technique, for any $t > 0$, we can write

$$\begin{aligned}
\mathbb{P}\left[S_m \geq \epsilon\right] &\leq e^{-t\epsilon}\, \mathbb{E}[e^{tS_m}] \\
&= e^{-t\epsilon}\, \mathbb{E}\left[e^{tS_{m-1}}\, \mathbb{E}[e^{tV_m}|X_1, \ldots, X_{m-1}]\right] \\
&\leq e^{-t\epsilon}\, \mathbb{E}[e^{tS_{m-1}}] e^{t^2 c_m^2/8} && \text{(lemma D.6)} \\
&\leq e^{-t\epsilon} e^{t^2 \sum_{i=1}^m c_i^2/8} && \text{(iterating previous argument)} \\
&= e^{-2\epsilon^2/\sum_{i=1}^m c_i^2}\,,
\end{aligned}$$

where we chose $t = 4\epsilon/\sum_{i=1}^m c_i^2$ to minimize the upper bound. This proves the first statement of the theorem, and the second statement is shown in a similar way. □

D.7 McDiarmid's inequality

The following is the main result of this section. Its proof makes use of Azuma's inequality.

Theorem D.8 (McDiarmid's inequality) *Let $X_1, \ldots, X_m \in \mathcal{X}^m$ be a set of $m \geq 1$ independent random variables and assume that there exist $c_1, \ldots, c_m > 0$ such that $f\colon \mathcal{X}^m \to \mathbb{R}$ satisfies the following conditions:*

$$\left|f(x_1, \ldots, x_i, \ldots, x_m) - f(x_1, \ldots, x_i', \ldots x_m)\right| \leq c_i\,, \tag{D.15}$$

for all $i \in [m]$ and any points $x_1, \ldots, x_m, x_i' \in \mathcal{X}$. Let $f(S)$ denote $f(X_1, \ldots, X_m)$, then, for all $\epsilon > 0$, the following inequalities hold:

$$\mathbb{P}[f(S) - \mathbb{E}[f(S)] \geq \epsilon] \leq \exp\left(\frac{-2\epsilon^2}{\sum_{i=1}^m c_i^2}\right) \tag{D.16}$$

$$\mathbb{P}[f(S) - \mathbb{E}[f(S)] \leq -\epsilon] \leq \exp\left(\frac{-2\epsilon^2}{\sum_{i=1}^m c_i^2}\right). \tag{D.17}$$

Proof: Define a sequence of random variables V_k, $k \in [m]$, as follows: $V = f(S) - \mathbb{E}[f(S)]$, $V_1 = \mathbb{E}[V|X_1] - \mathbb{E}[V]$, and for $k > 1$,

$$V_k = \mathbb{E}[V|X_1, \ldots, X_k] - \mathbb{E}[V|X_1, \ldots, X_{k-1}]\,.$$

Note that $V = \sum_{k=1}^m V_k$. Furthermore, the random variable $\mathbb{E}[V|X_1, \ldots, X_k]$ is a function of $X_1, \ldots, X_k$. Conditioning on $X_1, \ldots, X_{k-1}$ and taking its expectation is therefore:

$$\mathbb{E}\left[\mathbb{E}[V|X_1, \ldots, X_k]|X_1, \ldots, X_{k-1}\right] = \mathbb{E}[V|X_1, \ldots, X_{k-1}],$$

which implies $\mathbb{E}[V_k|X_1, \ldots, X_{k-1}] = 0$. Thus, the sequence $(V_k)_{k\in[m]}$ is a martingale difference sequence. Next, observe that, since $\mathbb{E}[f(S)]$ is a scalar, V_k can be expressed as follows:

$$V_k = \mathbb{E}[f(S)|X_1, \ldots, X_k] - \mathbb{E}[f(S)|X_1, \ldots, X_{k-1}]\,.$$

Thus, we can define an upper bound W_k and lower bound U_k for V_k by:

$$W_k = \sup_x \mathbb{E}[f(S)|X_1, \ldots, X_{k-1}, x] - \mathbb{E}[f(S)|X_1, \ldots, X_{k-1}]$$

$$U_k = \inf_x \mathbb{E}[f(S)|X_1, \ldots, X_{k-1}, x] - \mathbb{E}[f(S)|X_1, \ldots, X_{k-1}].$$

Now, by (D.15), for any $k \in [m]$, the following holds:

$$W_k - U_k = \sup_{x,x'} \mathbb{E}[f(S)|X_1, \ldots, X_{k-1}, x] - \mathbb{E}[f(S)|X_1, \ldots, X_{k-1}, x'] \leq c_k\,, \tag{D.18}$$

thus, $U_k \leq V_k \leq U_k + c_k$. In view of these inequalities, we can apply Azuma's inequality to $V = \sum_{k=1}^m V_k$, which yields exactly (D.16) and (D.17). □

McDiarmid's inequality is used in several of the proofs in this book. It can be understood in terms of stability: if changing any of its argument affects f only in a limited way, then, its deviations from its mean can be exponentially bounded. Note also that Hoeffding's in-

equality (theorem D.2) is a special instance of McDiarmid's inequality where f is defined by $f\colon (x_1, \ldots, x_m) \mapsto \frac{1}{m}\sum_{i=1}^m x_i$.

D.8 Normal distribution tails: Lower bound

If N is a random variable following the standard normal distribution, then for $u > 0$,

$$\mathbb{P}[N \geq u] \geq \frac{1}{2}\Big(1 - \sqrt{1 - e^{-u^2}}\Big). \tag{D.19}$$

D.9 Khintchine-Kahane inequality

The following inequality is useful in a variety of different contexts, including in the proof of a lower bound for the empirical Rademacher complexity of linear hypotheses (chapter 6).

Theorem D.9 (Khintchine-Kahane inequality) *Let $(\mathbb{H}, \|\cdot\|)$ be a normed vector space and let $\mathbf{x}_1, \ldots, \mathbf{x}_m$ be $m \geq 1$ elements of $\mathbb{H}$. Let $\boldsymbol{\sigma} = (\sigma_1, \ldots, \sigma_m)^\top$ with σ_is independent uniform random variables taking values in $\{-1, +1\}$ (Rademacher variables). Then, the following inequalities hold:*

$$\frac{1}{2}\mathop{\mathbb{E}}_{\boldsymbol{\sigma}}\Big[\Big\|\sum_{i=1}^m \sigma_i \mathbf{x}_i\Big\|^2\Big] \leq \Big(\mathop{\mathbb{E}}_{\boldsymbol{\sigma}}\Big[\Big\|\sum_{i=1}^m \sigma_i \mathbf{x}_i\Big\|\Big]\Big)^2 \leq \mathop{\mathbb{E}}_{\boldsymbol{\sigma}}\Big[\Big\|\sum_{i=1}^m \sigma_i \mathbf{x}_i\Big\|^2\Big]. \tag{D.20}$$

Proof: The second inequality is a direct consequence of the convexity of $x \mapsto x^2$ and Jensen's inequality (theorem B.20).

To prove the left-hand side inequality, first note that for any $\beta_1, \ldots, \beta_m \in \mathbb{R}$, expanding the product $\prod_{i=1}^m (1 + \beta_i)$ leads exactly to the sum of all monomials $\beta_1^{\delta_1} \cdots \beta_m^{\delta_m}$, with exponents $\delta_1, \ldots, \delta_m$ in $\{0, 1\}$. We will use the notation $\beta_1^{\delta_1} \cdots \beta_m^{\delta_m} = \beta^{\boldsymbol{\delta}}$ and $|\boldsymbol{\delta}| = \sum_{i=1}^m \delta_m$ for any $\boldsymbol{\delta} = (\delta_1, \ldots, \delta_m) \in \{0, 1\}^m$. In view of that, for any $(\alpha_1, \ldots, \alpha_m) \in \mathbb{R}^m$ and $t > 0$, the following equality holds:

$$t^2 \prod_{i=1}^m (1 + \alpha_i/t) = t^2 \sum_{\boldsymbol{\delta} \in \{0,1\}^m} \alpha^{\boldsymbol{\delta}}/t^{|\boldsymbol{\delta}|} = \sum_{\boldsymbol{\delta} \in \{0,1\}^m} t^{2-|\boldsymbol{\delta}|}\alpha^{\boldsymbol{\delta}}.$$

Differentiating both sides with respect to t and setting $t = 1$ yields

$$2\prod_{i=1}^m (1 + \alpha_i) - \sum_{j=1}^m \alpha_j \prod_{i \neq j}(1 + \alpha_i) = \sum_{\boldsymbol{\delta} \in \{0,1\}^m} (2 - |\boldsymbol{\delta}|)\alpha^{\boldsymbol{\delta}}. \tag{D.21}$$

For any $\boldsymbol{\sigma} \in \{-1, +1\}^m$, let $S_{\boldsymbol{\sigma}}$ be defined by $S_{\boldsymbol{\sigma}} = \|s_{\boldsymbol{\sigma}}\|$ with $s_{\boldsymbol{\sigma}} = \sum_{i=1}^m \sigma_i \mathbf{x}_i$. Then, setting $\alpha_i = \sigma_i \sigma'_i$, multiplying both sides of (D.21) by $S_{\boldsymbol{\sigma}} S_{\boldsymbol{\sigma}'}$, and taking the sum over all $\boldsymbol{\sigma}, \boldsymbol{\sigma}' \in \{-1, +1\}^m$ yields

$$\begin{aligned}
&\sum_{\boldsymbol{\sigma}, \boldsymbol{\sigma}' \in \{-1,+1\}^m} \Big(2\prod_{i=1}^m (1 + \sigma_i \sigma'_i) - \sum_{j=1}^m \sigma_j \sigma'_j \prod_{i \neq j}(1 + \sigma_i \sigma'_i)\Big) S_{\boldsymbol{\sigma}} S_{\boldsymbol{\sigma}'} \\
&\qquad = \sum_{\boldsymbol{\sigma}, \boldsymbol{\sigma}' \in \{-1,+1\}^m} \sum_{\boldsymbol{\delta} \in \{0,1\}^m} (2 - |\boldsymbol{\delta}|)\sigma^{\boldsymbol{\delta}} \sigma'^{\boldsymbol{\delta}} S_{\boldsymbol{\sigma}} S_{\boldsymbol{\sigma}'} \\
&\qquad = \sum_{\boldsymbol{\delta} \in \{0,1\}^m} (2 - |\boldsymbol{\delta}|) \sum_{\boldsymbol{\sigma}, \boldsymbol{\sigma}' \in \{-1,+1\}^m} \sigma^{\boldsymbol{\delta}} \sigma'^{\boldsymbol{\delta}} S_{\boldsymbol{\sigma}} S_{\boldsymbol{\sigma}'} \\
&\qquad = \sum_{\boldsymbol{\delta} \in \{0,1\}^m} (2 - |\boldsymbol{\delta}|)\Big[\sum_{\boldsymbol{\sigma} \in \{-1,+1\}^m} \sigma^{\boldsymbol{\delta}} S_{\boldsymbol{\sigma}}\Big]^2.
\end{aligned} \tag{D.22}$$

Note that the terms of the right-hand sum with $|\boldsymbol{\delta}| \geq 2$ are non-positive. The terms with $|\boldsymbol{\delta}| = 1$ are null: since $S_{\boldsymbol{\sigma}} = S_{-\boldsymbol{\sigma}}$, we have $\sum_{\boldsymbol{\sigma} \in \{-1,+1\}^m} \sigma^{\boldsymbol{\delta}} S_{\boldsymbol{\sigma}} = 0$ in that case. Thus, the right-hand side can be upper bounded by the term with $\boldsymbol{\delta} = 0$, that is, $2\Big(\sum_{\boldsymbol{\sigma} \in \{-1,+1\}^m} S_{\boldsymbol{\sigma}}\Big)^2$. The left-hand

side of (D.22) can be rewritten as follows:

$$\sum_{\boldsymbol{\sigma}\in\{-1,+1\}^m}(2^{m+1}-m2^{m-1})S_{\boldsymbol{\sigma}}^2+2^{m-1}\sum_{\substack{\boldsymbol{\sigma}\in\{-1,+1\}^m\\ \boldsymbol{\sigma}'\in B(\boldsymbol{\sigma},1)}}S_{\boldsymbol{\sigma}}S_{\boldsymbol{\sigma}'}$$

$$=2^m\sum_{\boldsymbol{\sigma}\in\{-1,+1\}^m}S_{\boldsymbol{\sigma}}^2+2^{m-1}\sum_{\boldsymbol{\sigma}\in\{-1,+1\}^m}S_{\boldsymbol{\sigma}}\Bigg(\sum_{\boldsymbol{\sigma}'\in B(\boldsymbol{\sigma},1)}S_{\boldsymbol{\sigma}'}-(m-2)S_{\boldsymbol{\sigma}}\Bigg), \quad \text{(D.23)}$$

where $B(\boldsymbol{\sigma},1)$ denotes the set of $\boldsymbol{\sigma}'$ that differ from $\boldsymbol{\sigma}$ in exactly one coordinate $j\in[m]$, that is the set of $\boldsymbol{\sigma}'$ with Hamming distance one from $\boldsymbol{\sigma}$. Note that for any such $\boldsymbol{\sigma}'$, $s_{\boldsymbol{\sigma}}-s_{\boldsymbol{\sigma}'}=2\sigma_j\mathbf{x}_j$ for one coordinate $j\in[m]$, thus, $\sum_{\boldsymbol{\sigma}'\in B(\boldsymbol{\sigma},1)}s_{\boldsymbol{\sigma}}-s_{\boldsymbol{\sigma}'}=2s_{\boldsymbol{\sigma}}$. In light of that and using the triangle inequality, we can write

$$\begin{aligned}(m-2)S_{\boldsymbol{\sigma}}=\|ms_{\boldsymbol{\sigma}}\|-\|2s_{\boldsymbol{\sigma}}\|&=\Big\|\sum_{\boldsymbol{\sigma}'\in B(\boldsymbol{\sigma},1)}s_{\boldsymbol{\sigma}}\Big\|-\Big\|\sum_{\boldsymbol{\sigma}'\in B(\boldsymbol{\sigma},1)}s_{\boldsymbol{\sigma}}-s_{\boldsymbol{\sigma}'}\Big\|\\&\le\Big\|\sum_{\boldsymbol{\sigma}'\in B(\boldsymbol{\sigma},1)}s_{\boldsymbol{\sigma}'}\Big\|\le\sum_{\boldsymbol{\sigma}'\in B(\boldsymbol{\sigma},1)}S_{\boldsymbol{\sigma}'}.\end{aligned}$$

Thus, the second sum of (D.23) is non-negative and the left-hand side of (D.22) can be lower bounded by the first sum $2^m\sum_{\boldsymbol{\sigma}\in\{-1,+1\}^m}S_{\boldsymbol{\sigma}}^2$. Combining this with the upper bound found for (D.22) gives

$$2^m\sum_{\boldsymbol{\sigma}\in\{-1,+1\}^m}S_{\boldsymbol{\sigma}}^2\le 2\Big[\sum_{\boldsymbol{\sigma}\in\{-1,+1\}^m}S_{\boldsymbol{\sigma}}\Big]^2.$$

Dividing both sides by 2^{2m} and using $\mathbb{P}[\boldsymbol{\sigma}]=1/2^m$ gives $\mathbb{E}_{\boldsymbol{\sigma}}[S_{\boldsymbol{\sigma}}^2]\le 2(\mathbb{E}_{\boldsymbol{\sigma}}[S_{\boldsymbol{\sigma}}])^2$ and completes the proof. □

The constant 1/2 appearing in the first inequality of (D.20) is optimal. To see this, consider the case where $m=2$ and $\mathbf{x}_1=\mathbf{x}_2=\mathbf{x}$ for some non-zero vector $\mathbf{x}\in\mathbb{H}$. Then, the left-hand side of the first inequality is $\frac{1}{2}\sum_{i=1}^m\|\mathbf{x}_i\|^2=\|\mathbf{x}\|^2$ and the right-hand side $\big(\mathbb{E}_{\boldsymbol{\sigma}}\left[\|(\sigma_1+\sigma_2)\mathbf{x}\|\right]\big)^2=\|\mathbf{x}\|^2(\mathbb{E}_{\boldsymbol{\sigma}}[|\sigma_1+\sigma_2|])^2=\|\mathbf{x}\|^2$.

Note that when the norm $\|\cdot\|$ corresponds to an inner product, as in the case of a Hilbert space $\mathbb{H}$, we can write

$$\mathop{\mathbb{E}}_{\boldsymbol{\sigma}}\Big[\Big\|\sum_{i=1}^m\sigma_i\mathbf{x}_i\Big\|^2\Big]=\sum_{i,j=1}^m\mathop{\mathbb{E}}_{\boldsymbol{\sigma}}\Big[\sigma_i\sigma_j(\mathbf{x}_i\cdot\mathbf{x}_j)\Big]=\sum_{i,j=1}^m\mathop{\mathbb{E}}_{\boldsymbol{\sigma}}[\sigma_i\sigma_j](\mathbf{x}_i\cdot\mathbf{x}_j)=\sum_{i=1}^m\|\mathbf{x}_i\|^2,$$

since by the independence of the random variables σ_i, for $i\ne j$, $\mathbb{E}_{\boldsymbol{\sigma}}[\sigma_i\sigma_j]=\mathbb{E}_{\boldsymbol{\sigma}}[\sigma_i]\,\mathbb{E}_{\boldsymbol{\sigma}}[\sigma_j]=0$. Thus, (D.20) can then be rewritten as follows:

$$\frac{1}{2}\sum_{i=1}^m\|\mathbf{x}_i\|^2\le\Big(\mathop{\mathbb{E}}_{\boldsymbol{\sigma}}\Big[\Big\|\sum_{i=1}^m\sigma_i\mathbf{x}_i\Big\|\Big]\Big)^2\le\sum_{i=1}^m\|\mathbf{x}_i\|^2. \quad \text{(D.24)}$$

D.10 Maximal inequality

The following gives an upper bound on the expectation of the maximum of a finite set of random variables that is useful in several contexts.

Theorem D.10 (Maximal inequality) *Let $X_1\dots X_n$ be $n\ge 1$ real-valued random variables such that for all $j\in[n]$ and $t>0$, $\mathbb{E}[e^{tX_j}]\le e^{\frac{t^2r^2}{2}}$ for some $r>0$. Then, the following inequality holds:*

$$\mathbb{E}\Big[\max_{j\in[n]}X_j\Big]\le r\sqrt{2\log n}.$$

Proof: For any $t>0$, by the convexity of exp and Jensen's inequality, the following holds:

$$e^{t\,\mathbb{E}[\max_{j\in[n]}X_j]}\le\mathbb{E}[e^{t\max_{j\in[n]}X_j}]=\mathbb{E}\Big[\max_{j\in[n]}e^{tX_j}\Big]\le\mathbb{E}\Big[\sum_{j\in[n]}e^{tX_j}\Big]\le ne^{\frac{t^2r^2}{2}}.$$

Taking the log of both sides yields

$$\mathbb{E}\left[\max_{j\in[n]} X_j\right] \leq \frac{\log n}{t} + \frac{tr^2}{2}. \tag{D.25}$$

Choosing $t = \frac{\sqrt{2\log n}}{r}$, which minimizes the right-hand side, gives the upper bound $r\sqrt{2\log n}$. □

Note that, in view of the expression of their moment-generating function (equation (C.24)), for standard Gaussian random variables X_j, the assumptions of the theorem hold as equalities: $\mathbb{E}[e^{tX_j}] = e^{\frac{t^2}{2}}$.

Corollary D.11 (Maximal inequality) *Let $X_1 \dots X_n$ be $n \geq 1$ real-valued random variables such that for all $j \in [n]$, $X_j = \sum_{i=1}^m Y_{ij}$ where, for each fixed $j \in [n]$, Y_{ij} are independent zero mean random variables taking values in $[-r_i, +r_i]$, for some $r_i > 0$. Then, the following inequality holds:*

$$\mathbb{E}\left[\max_{j\in[n]} X_j\right] \leq r\sqrt{2\log n},$$

with $r = \sqrt{\sum_{i=1}^m r_i^2}$.

Proof: By the independence of the Y_{ij}s for fixed j and Hoeffding's lemma (Lemma D.1), the following inequality holds for all $j \in [n]$:

$$\mathbb{E}[e^{tX_j}] = \mathbb{E}\left[\prod_{i=1}^m e^{tY_{ij}}\right] = \prod_{i=1}^m \mathbb{E}[e^{tY_{ij}}] \leq \prod_{i=1}^m e^{\frac{t^2 r_j^2}{2}} = e^{\frac{t^2 r^2}{2}}, \tag{D.26}$$

with $r^2 = \sum_{i=1}^m r_i^2$. The result then follows immediately by Theorem D.10. □

D.11 Chapter notes

Several of the concentration inequalities presented in this chapter are based on a bounding technique due to Chernoff [1952]. Theorem D.3 is due to Sanov [1957]. For the exponential inequality of exercise D.7, which is an alternative form of Sanov's inequality, see [Hagerup and Rüb, 1990] and the references therein. The multiplicative Chernoff bounds presented in this chapter (Theorem D.4) were given by Angluin and Valiant [1979]. Hoeffding's inequality and lemma (Lemma D.1 and Theorem D.2) are due to Hoeffding [1963]. The improved version of Azuma's inequality [Hoeffding, 1963, Azuma, 1967] presented in this chapter is due to McDiarmid [1989]. The improvement is a reduction of the exponent by a factor of 4. This also appears in McDiarmid's inequality, which is derived from the inequality for bounded martingale sequences. The inequalities presented in exercise D.6 are due to Bernstein [1927] and Bennett [1962]; the exercise is from Devroye and Lugosi [1995].

The binomial inequality of section D.5 is due to Slud [1977]. The tail bound of section D.8 is due to Tate [1953] (see also Anthony and Bartlett [1999]). The Khintchine-Kahane inequality was first studied in the case of real-valued variables $x_1, \dots, x_m$ by Khintchine [1923], with better constants and simpler proofs later provided by Szarek [1976], Haagerup [1982], and Tomaszewski [1982]. The inequality was extended to normed vector spaces by Kahane [1964]. The proof presented here is due to Latała and Oleszkiewicz [1994] and provides the best possible constants.

D.12 Exercises

D.1 Twins paradox. Professor Mamoru teaches at a university whose computer science and math building has $F = 30$ floors.

(1) Assume that the floors are independent and that they have the same probability to be selected by someone taking the elevator. How many people should take the elevator in order to make it likely (probability more than half) that two of them go to the same floor?

(*Hint*: use the Taylor series expansion of $e^{-x} = 1 - x + \ldots$ and give an approximate general expression of the solution.)

(2) Professor Mamoru is popular, and his floor is in fact more likely to be selected than others. Assuming that all other floors are equiprobable, derive the general expression of the probability that two people go to the same floor, using the same approximation as before. How many people should take the elevator in order to make it likely that two of them go to the same floor when the probability of Professor Mamoru's floor is .25, .35, or .5? When $q = .5$, would the answer change if the number of floors were instead $F = 1{,}000$?

(3) The probability models assumed in (1) and (2) are both naive. If you had access to the data collected by the elevator guard, how would you define a more faithful model?

D.2 Estimating label bias. Let $\mathcal{D}$ be a distribution over $\mathcal{X}$ and let $f\colon \mathcal{X} \times \{-1, +1\}$ be a labeling function. Suppose we wish to find a good approximation of the label bias of the distribution $\mathcal{D}$, that is of p_+ defined by:

$$p_+ = \mathop{\mathbb{P}}_{x \sim \mathcal{D}}[f(x) = +1]. \tag{D.27}$$

Let $\mathcal{S}$ be a finite labeled sample of size m drawn i.i.d. according to $\mathcal{D}$. Use $\mathcal{S}$ to derive an estimate $\widehat{p}_+$ of p_+. Show that for any $\delta > 0$, with probability at least $1 - \delta$, $|p_+ - \widehat{p}_+| \leq \sqrt{\frac{\log(2/\delta)}{2m}}$.

D.3 Biased coins. Professor Moent has two coins in his pocket, coin x_A and coin x_B. Both coins are slightly biased, i.e., $\mathbb{P}[x_A = 0] = 1/2 - \epsilon/2$ and $\mathbb{P}[x_B = 0] = 1/2 + \epsilon/2$, where $0 < \epsilon < 1$ is a small positive number, 0 denotes heads and 1 denotes tails. He likes to play the following game with his students. He picks a coin $x \in \{x_A, x_B\}$ from his pocket uniformly at random, tosses it m times, reveals the sequence of 0s and 1s he obtained and asks which coin was tossed. Determine how large m needs to be for a student's coin prediction error to be at most $\delta > 0$.

(a) Let S be a sample of size m. Professor Moent's best student, Oskar, plays according to the decision rule $f_o\colon \{0,1\}^m \to \{x_A, x_B\}$ defined by $f_o(S) = x_A$ iff $N(S) < m/2$, where $N(S)$ is the number of 0's in sample S.

Suppose m is even, then show that

$$error(f_o) \geq \frac{1}{2}\,\mathbb{P}\left[N(S) \geq \frac{m}{2} \middle| x = x_A\right]. \tag{D.28}$$

(b) Assuming m even, show that

$$error(f_o) > \frac{1}{4}\left[1 - \left[1 - e^{-\frac{m\epsilon^2}{1-\epsilon^2}}\right]^{\frac{1}{2}}\right]. \tag{D.29}$$

(c) Argue that if m is odd, the probability can be lower bounded by using $m+1$ in the bound in (a) and conclude that for both odd and even m,

$$error(f_o) > \frac{1}{4}\left[1 - \left[1 - e^{-\frac{2\lceil m/2\rceil\epsilon^2}{1-\epsilon^2}}\right]^{\frac{1}{2}}\right]. \tag{D.30}$$

(d) Using this bound, how large must m be if Oskar's error is at most δ, where $0 < \delta < 1/4$. What is the asymptotic behavior of this lower bound as a function of ϵ?

(e) Show that no decision rule $f\colon \{0,1\}^m \to \{x_A, x_B\}$ can do better than Oskar's rule f_o. Conclude that the lower bound of the previous question applies to all rules.

D.4 Concentration bounds. Let X be a non-negative random variable satisfying $\mathbb{P}[X > t] \leq ce^{-2mt^2}$ for all $t > 0$ and some $c > 0$. Show that $\mathbb{E}[X^2] \leq \frac{\log(ce)}{2m}$ (*Hint*: to do that, use the

identity $\mathbb{E}[X^2] = \int_0^{+\infty} \mathbb{P}[X^2 > t]dt$, write $\int_0^{+\infty} = \int_0^u + \int_u^{+\infty}$, bound the first term by u and find the best u to minimize the upper bound).

D.5 Comparison of Hoeffding's and Chebyshev's inequalities. Let $X_1, \ldots, X_m$ be a sequence of random variables taking values in $[0, 1]$ with the same mean μ and variance $\sigma^2 < \infty$ and let $\overline{X} = \frac{1}{m}\sum_{i=1}^m X_i$.

(a) For any $\epsilon > 0$, give a bound on $\mathbb{P}[|\overline{X} - \mu| > \epsilon]$ using Chebyshev's inequality, then Hoeffding's inequality. For what values of σ is Chebyshev's inequality tighter?

(b) Assume that the random variables X_i take values in $\{0, 1\}$. Show that $\sigma^2 \leq \frac{1}{4}$. Use this to simplify Chebyshev's inequality. Choose $\epsilon = .05$ and plot Chebyshev's inequality thereby modified and Hoeffding's inequality as a function of m (you can use your preferred program for generating the plots).

D.6 Bennett's and Bernstein's inequalities. The objective of this problem is to prove these two inequalities.

(a) Show that for any $t > 0$, and any random variable X with $\mathbb{E}[X] = 0$, $\mathbb{E}[X^2] = \sigma^2$, and $X \leq c$,

$$\mathbb{E}[e^{tX}] \leq e^{f(\sigma^2/c^2)}, \tag{D.31}$$

where

$$f(x) = \log\Big(\frac{1}{1+x}e^{-ctx} + \frac{x}{1+x}e^{ct}\Big).$$

(b) Show that $f''(x) \leq 0$ for $x \geq 0$.

(c) Using Chernoff's bounding technique, show that

$$\mathbb{P}\left[\frac{1}{m}\sum_{i=1}^m X_i \geq \epsilon\right] \leq e^{-tm\epsilon + \sum_{i=1}^m f(\sigma^2_{X_i}/c^2)},$$

where ($\sigma^2_{X_i}$ is the variance of X_i.

(d) Show that $f(x) \leq f(0) + xf'(0) = (e^{ct} - 1 - ct)x$.

(e) Using the bound derived in (4), find the optimal value of t.

(f) *Bennett's inequality*. Let $X_1, \ldots, X_m$ be independent real-valued random variables with zero mean such that for $i = 1, \ldots, m$, $X_i \leq c$. Let $\sigma^2 = \frac{1}{m}\sum_{i=1}^m \sigma^2_{X_i}$. Show that

$$\mathbb{P}\left[\frac{1}{m}\sum_{i=1}^m X_i > \epsilon\right] \leq \exp\left(-\frac{m\sigma^2}{c^2}\theta\left(\frac{\epsilon c}{\sigma^2}\right)\right), \tag{D.32}$$

where $\theta(x) = (1 + x)\log(1 + x) - x$.

(g) *Bernstein's inequality*. Show that under the same conditions as Bennett's inequality

$$\mathbb{P}\left[\frac{1}{m}\sum_{i=1}^m X_i > \epsilon\right] \leq \exp\left(-\frac{m\epsilon^2}{2\sigma^2 + 2c\epsilon/3}\right). \tag{D.33}$$

(*Hint*: show that for all $x \geq 0$, $\theta(x) \geq h(x) = \frac{3}{2}\frac{x^2}{x+3}$.)

(h) Write Hoeffding's inequality assuming the same conditions. For what values of σ is Bernstein's inequality better than Hoeffding's inequality?

D.7 Exponential inequality. Let X be a random variable following a binomial distribution $B(m, p)$.

(a) Use Sanov's inequality to show that the following *exponential inequality* holds for any $\epsilon > 0$:

$$\mathbb{P}\left[\frac{X}{m} - p > \epsilon\right] \leq \left[\left(\frac{p}{p+\epsilon}\right)^{p+\epsilon} \left(\frac{1-p}{1-(p+\epsilon)}\right)^{1-(p+\epsilon)}\right]^m. \tag{D.34}$$

(b) Use that to show that the following holds:

$$\mathbb{P}\left[\frac{X}{m} - p > \epsilon\right] \leq \left(\frac{p}{p+\epsilon}\right)^{m(p+\epsilon)} e^{m\epsilon}. \tag{D.35}$$

(c) Prove that

$$\mathbb{P}\left[\frac{X}{m} - p > \epsilon\right] \leq e^{-mp\theta(\frac{\epsilon}{p})}, \tag{D.36}$$

where θ is defined as in exercise D.6.